SECTION A · PS COMPLIANCE

A. The prototype against the PS, clause by clause

PS verbatim (PRD §2.1): "The GUI based model will take .IQ or .wav file as input and: (i) identify signal parameters (Sampling frequency, Modulation, FEC, Interleaving); (ii) demodulate signals (FSK, QAM, PSK); (iii) de-interleave (Block, Convolution, Diagonal, Pseudo Random); (iv) FEC (short-constraint convolutional codes with Viterbi decoding, RS block codes, Concatenated codes, LDPC); (v) bit stream correlation." Expected solution adds GUI feature visibility (sampling frequency, constellation, waterfall, demod, de-interleave, error correction, header/payload). Statuses verified in code on 28 Sep 2026 (recipe.py, search.py, kernels/, classify.py), not copied from any plan.

A.1 Input, GUI and spectral features — the "expected solution" half of the PS

PS item	Status	Evidence in the prototype
.IQ / .wav input	**PASS**	engine io: WAV (mono/stereo, int/float), raw IQ dtype sniffing, SigMF cf32/ci16; hand-verified in Chrome end-to-end
GUI model	**PASS**	React UI on a local FastAPI sidecar (sanjaya serve, token-authed, 127.0.0.1); upload, sample library, autopilot; verified in Chrome
Feature visibility: waterfall / PSD / constellation	**PASS**	Waterfall + PSD tiles + constellation from engine results; 100 MB interactive LOD budget measured feasible at 0.62 s (3× margin)
Sampling frequency identification	**PASS**	Header/SigMF; Sensor-Twin clock-ppm cross-checks on stitched captures; no-Fs case handled with normalised units
Automated analysis	**PASS**	One-click autopilot: upload → blind search → proof → signed certificate; CLI (gen/analyze/verify) on the same pipeline
Header / payload identification	**PASS**	Sync marker, frame period, counter/length fields, CRC polynomial + init solved by GF(2) gcd; header/payload split shown in UI
Trials on real signals	**PASS**	6 real recordings (STANAG 4285, MIL-188-110A, HFDL, PSK31, SITOR-B, 2G ALE) identify correctly: rates to 0.1 Bd, ppm-scale clock offsets

A.2 The five literal clauses (i)–(v)

Clause	Status	What works today	What does not (yet)
(i) Sampling frequency	**PASS**	From headers/SigMF; cross-checks via standards' symbol clocks and stitching	—
(i) Modulation ID	**PARTIAL**	BPSK/QPSK/8PSK/16QAM/64QAM via decode-fit; FSK family via tones; 6 real files identified	Blind search accepts only BPSK/QPSK as coding-search hypotheses; 8PSK/QAM blind-decode deliberately withheld (false-certificate risk)
(i) FEC ID	**PASS (known-recipe)**	Conv K3–9 (parity test), RS, concatenated, LDPC all certified by the proof with a supplied recipe	Not discoverable blind except conv (rate-1/2) — RS/LDPC/concat blind ID gated
(i) Interleaving ID	**PARTIAL**	Block geometry solved blind; diagonal/random invert known-recipe	Convolutional (Forney) interleaver absent from recipe schema entirely
(ii) Demodulate PSK	**PASS**	BPSK/QPSK/8PSK: feedforward timing (Oerder-Meyr), Viterbi-Viterbi phase, Gray LLR demap; QPSK 40/40 frames 0 errors at 8–12 dB; 6 real HF files	—
(ii) Demodulate QAM	**PASS (known-recipe)**	16QAM/64QAM: decision-directed amplitude rings, verified 40/40 CRC frames at 20/26 dB in known-recipe mode	Blind discovery not attempted (proof-safety, per search.py design note)
(ii) Demodulate FSK	**NOT BUILT**	FSK tones measured (SITOR-B 170 Hz shift, ALE 8×250 Hz plan) and identified	No FSK demodulator: different receiver architecture (tone-bank/non-coherent), explicitly deferred — the single largest gap
(iii) Block	**PASS**	Blind geometry solve; bit-exact interleave/de-interleave round trips	—

Clause	Status	What works today	What does not (yet)
(iii) Diagonal	**PASS (known-recipe)**	coding/diagonal_perm: helical cyclic-shift operator behind generic interleave()	Not blind-reachable (same scope boundary as block's siblings)
(iii) Pseudo-random	**PASS (known-recipe)**	coding/random_perm: seeded permutation, seed carried in recipe	Blind seed search not built
(iii) Convolutional	**NOT BUILT**	—	Absent: no Forney B/J interleaver in code; PS names it, so counted as a literal miss
(iv) Conv + Viterbi	**PASS**	Soft-decision Viterbi K3–9 NumPy fallback (0.15 Mbit/s, 0 errors @ 6 dB); blind K search; C++ kernels blocked on 64-bit toolchain	C++ speed target (R-09) unmeasured — fallback only
(iv) RS block	**PASS**	GF(2^m) BM+Chien+Forney, RS(15,9)/RS(63,51)/RS(255,223) tested; certifies via RS1/RS2 proof tests (Forney magnitude bug caught by 2-error test, fixed)	Blind (n,k) search not built
(iv) Concatenated	**PASS (known-recipe)**	RS outer + conv inner through the same chain and proof (K1/K2 on the inner code)	Blind peeling not built
(iv) LDPC	**PASS (known-recipe)**	Systematic GF(2) generator + log-domain min-sum BP; proof marks LDPC2 explicitly uncalibrated rather than dressing it up	Blind catalogue match not built; thresholds v0-defaults until G2 calibration
(v) Bit-stream correlation	**PASS**	Sync cross-correlation, counters, length fields, CRC solve (poly by gcd, init by inverse), header/payload labelling; S1–S4 proof tests	—

A.3 Scoreboard and the two literal misses

PS sub-item inventory	Count	Detail
Literal sub-items in clauses (i)–(v)	21	Fs · mod ID · FEC ID · interleave ID · demod ×3 · de-interleave ×4 · FEC decode ×4 · correlation (counting QAM and FSK demod separately and each named family once)
Working today, known-recipe path	19	Everything except FSK demodulation and the convolutional interleaver
Also reachable blind (no recipe supplied)	11–14	Fs (headers), BPSK/QPSK mod, conv K3–9 FEC, block interleaver, scrambler, sync, counters/CRC, header/payload; ± diagonal/random seeds and concatenated depending on how the same primitives are counted
Literal misses	2	(ii) FSK demod — no tone-bank receiver, largest remaining gap, own TX/RX chain needed; (iii) convolutional interleaver — absent from the schema; both louder-than-silent: an FSK recipe cannot be built, a conv-interleaver recipe cannot be expressed
UI gaps vs the PRD's own bar (not the PS text)	3	Eye/FSK views still demo drawings; multi-emitter Scene and genome D5 not built; job worker is a thread, not the PRD's per-job process with a memory cap

Bottom line on "is the prototype done as the PS says?"

Yes to the shape of the PS: GUI in, .IQ/.wav in, automated analysis, spectral features, header/payload out — all working and demonstrated end-to-end, including on real off-air recordings. Yes to 19 of 21 named techniques in the honest, recipe-supplied sense the PS never distinguishes but every engineer will. No on two literal items (FSK demod, convolutional interleaver), which is the right thing to fix next: FSK is the one the PS names first among demodulation families, and 28 of 44 public repos already claim it on paper. One caution for the finale: the PRD's KPIs (BER within 0.5 dB of theory, ppm bounds, decision rates) are pre-registered targets, not yet measured — claim coverage and working code, not KPI compliance.

A.4 Read-only discipline observed for this audit

- No project file created, modified or deleted: code inspected via read/grep only; the competitor snapshot (_build/data.json) read in place; no repository cloned, pulled or executed; tests counted statically (15 files, 52 named functions plus parametrised cases) instead of running pytest, because a pytest run would drop .pytest_cache into the tree.

- git status after the audit shows exactly one addition by this audit: 147prototype_review.pdf in the project root. All other working-tree changes predate this task and belong to the build sessions (committed history through 4cec7bd).

SECTION B · INNOVATION ASSESSMENT

B. What is genuinely new here, against every public repo for this PS

Source: _build/data.json, a 26 Sep 2026 snapshot of 52 GitHub repositories matching SIH 2026 PS 26147 — READMEs, file counts, commit counts, languages, push dates. Nothing was cloned or run: the comparison reads what teams say about their own work, which is exactly what a judge sees before the demo. 44 remain after hygiene exclusions (1 off-topic 2020 archive, 2 duplicate forks, 3 forks of other entries, 3 docs-only with zero code files).

B.1 What the 44-repo field actually contains (re-derived from the snapshot, reproducible)

Capability claimed in READMEs	Repos	Reading
Demodulate PSK/QAM/FSK	32–32–28	The crowded half of the PS. 6 repos claim the full breadth matrix: FSK + LDPC + convolutional interleaver + pseudo-random (harikesh2709 spectra-signal-analyzer 286 files, Venkata-Manoj RF-signal-analysis 278 files/51 commits, yoganandasp15 164 files, Devansh-567 95, WaghJagdish 91, ShantanuSomwanshi 51)
Reed-Solomon / LDPC	23 / 22	Named on paper widely; few show decode-quality evidence
"Tests" mentioned	31	Mostly "we tested it"; few publish runnable suites
SigMF mentioned	12	A quarter of the field; mostly as an input format
Real named signals (STANAG/NAVTEX/HFDL/…)	≈4	The field is overwhelmingly synthetic-only: truth comes from the team's own generator
GNU Radio mentioned	15	Rarely load-bearing: usually a topic tag or an export promise
Scrambler identification	4	Almost nobody touches the scrambler layer
Data-sufficiency ("enough data?")	7	A sentence or a dashboard widget; no predictions that are then verified
Multi-capture stitching	1	Mentioned once, in passing, by one repo
Re-checkable verification	0	See B.2 — the decisive number

B.2 The verification gap, keyword by keyword "Verify" is the most misused word in the field. It usually means we checked our output — not you can re-check our output without trusting us.

Keyword in 44 READMEs	Hits	What the hits actually are, on reading
"verify / verification"	30	Merging hunks, lint gates, "verified against ground truth", result windows that say VERIFIED — none re-checkable by a third party
"tests"	31	"We tested it" sentences; few runnable suites; nearly none failure-proofing wrong-hypothesis behaviour
CRC mentioned	9	Validation inside the pipeline (fine and common). Closest thing to a gate: Gururaghavendra123 publish a decode only when CRC passes
"certificate"/"signature"/Ed25519	9	Generic words; zero signing, zero third-party re-verification anywhere in the field
A mechanism a stranger can re-run	0	No repo in the snapshot publishes anything a second person can re-execute against the file to re-produce the verdict

B.3 The five structural gaps, mapped to what SANJAYA has already shipped

What no public repo does	Field status	SANJAYA today	Why it holds
C1 · Layered proof of decode + Ed25519-signed certificate, re-checkable via sanjaya verify / the /verify page	0/44	Working end-to-end: PHY T1–T4 vs measured noise reference, coding K1–K3, structure S1–S4; ~1.7 s re-check; tampered-certificate and wrong-file cases are pinned by tests	Shows why, not just what: K1 separates the true code from random bits by ≈88σ (0.0003 vs 0.494) while the waveform alone cannot (0.002 vs 0.001 dB at 12 dB)

What no public repo does	Field status	SANJAYA today	Why it holds
C2 · Data-Sufficiency Engine: predicts the data needed, then verifies the prediction	7/44 keyword only	Ledger of what limited each decision; frame-count prediction verified by truncation test (a capture cut to the P90 prediction decodes); power analysis separates "no code exists" from "not enough data"	Changes the collector's workflow; nobody else answers "is this enough data?" with a number
C3 · Evidence stitching across sensors/captures + Sensor Twin	1/44 keyword only	Three-capture demo (76.8/61.152/96 kHz, 0/−25/+40 ppm, one inverted) pools, decodes, catches a planted decoy, returns −25.6/+37.4 ppm	Directly answers the PS's own motivation: recordings "from different sensors and different locations"
C4 · Recipe Compiler: a dependency-free receiver emitted per decode	0/44	receiver.zip per decoded job; bit-exactness gate re-runs the receiver on the source recording before release; portable test passed on a different ADC rate/offset/inversion	Answers the PS's "output utilised in the designated sensors"; turns a certificate into an artefact
C5 · Scrambler-layer recovery	4/44 keywords, no mechanism	Blind additive-scrambler solve incl. post-FEC cosets, degree ≤ 8 exhaustive + library	Even strong repos stop at the code layer

Innovation verdict

Measured against every public repo findable for this PS, SANJAYA is not an incremental improvement on the field — it plays a different game. The field competes on breadth of demodulation claims; nobody has any re-checkable verification (0/44), nobody answers "is this enough data?" with a verified prediction, one repo even mentions stitching, nobody emits a receiver, and the scrambler layer is almost untouched. All five of those are working SANJAYA features with tests, not roadmap items. The most defensible claim at the finale: "pick any public repo for this PS; you cannot re-check its answer, and it cannot tell you when it does not know. Ours you can — in 1.7 seconds, from the file alone." Breadth-on-paper is the field's commodity; proof-of-decode is the moat, and the moat's parts interlock (the compiler only ships proof-gated recipes; sufficiency needs the proof's evidence ledger; stitching needs the Twin), so copying one piece does not reproduce the system.

B.4 Where we are honestly behind — read this before the pitch, not after

Dimension	Reality	Consequence
Shipped blind breadth	Several repos claim working FSK/8PSK/LDPC decode demos; our blind search covers BPSK/QPSK + conv + block + scrambler only. 8PSK/QAM decode needs a supplied recipe.	A judge who reads READMEs first will expect one-click FSK decode. Fix by closing FSK (Part C), then let the certificate's layered proof be the differentiator on everything else
Commit momentum	Field leaders: 100 commits (RadioFry), 51 (Venkata-Manoj), 40 (arachknight66); ours: 13.	Visible engineering activity is a judging signal. Cheap to fix: commit at feature granularity
Stars / traction	One docs-only repo holds 30 stars; working repos mostly 0–2.	Cosmetic; ignore
Measured KPIs	PRD targets (BER ± 0.5 dB of theory, ppm bounds) are pre-registered, not yet bench-measured.	Never present a target as a result; the certificate itself models the honesty the team sells

SECTION C · CLOSING PLAN

C. What it takes to say "done as the PS says", in order

Priority by leverage on the PS text and on what judges can see. Items 1–2 change the compliance answer; the rest harden it.

#	Action	Why this order	Scope
1	FSK demodulation (2/4/8-FSK): tone-bank TX + non-coherent tone-bank RX + FSK LLRs feeding the existing FEC layer	The last named demodulation family; converts the biggest literal miss into a PASS; real FSK files (SITOR-B, ALE, RTTY, HF DSC) are already correctly identified by classify.py, so the demo only needs the decode half	Own TX/RX chain, one focused session
2	Convolutional (Forney B/J) interleaver: operator in coding/, kind in the recipe schema, invertibility test	The second literal miss; small, well-defined, removes the only other NOT BUILT	Half a session

#	Action	Why this order	Scope
3	Modulation-aware blind search: generalise the 2-bit/symbol dual-code model or add a modulation-hypothesis outer loop driven by classify.py	Turns the existing BPSK/8PSK/16QAM/64QAM demod capability into blind PS-clause wins; needs the same care BPSK got (one family at a time, no unverified hypotheses)	Careful pass, next after 1–2
4	RS/LDPC/concat + diagonal/random into blind reachability	PS breadth inside the blind path; primitives already exist and certify known-recipe	Follows 3
5	Bench-calibrate the proof thresholds (RS1/RS2, LDPC1/2, corrected K1/K2) and record KPI measurements	Moves claims from "works" to "meets the pre-registered KPIs"; needed before any number is spoken at the finale	G2 work as planned

Final word

Compliance: 19/21 literal items working, 2 named and honestly missing, UI demonstrably live on real recordings — a strong "almost done" that a two-item closing plan finishes. Innovation: five shipped capabilities that the entire 44-repo public field lacks, anchored by the only re-checkable verification mechanism in the PS's ecosystem. The remaining risk is not novelty or coverage — it is the temptation to claim breadth-on-paper like everyone else. The prototype's whole argument is that it does not have to.